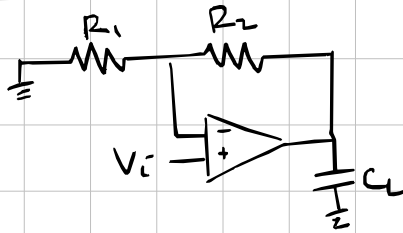
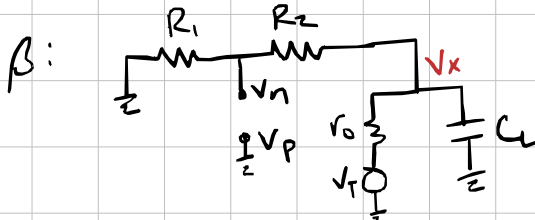


AEC HW9 (

①



$$f_c = 10 \text{ MHz}, r_o = 75 \Omega, r_d = \infty$$



$$V_n = V_x \frac{R_1}{R_1 + R_2}$$

$$V_x = V_T (R_1 + R_2) \parallel C_L / (R_1 + R_2) \parallel C_L + r_o$$

$$(R_1 + R_2) \parallel C_L = (R_1 + R_2) / (j\omega C_L) / [(R_1 + R_2) + 1/j\omega C_L]$$

$$= (R_1 + R_2) / (1 + j\omega (R_1 + R_2) C_L)$$

$$V_x = V_T [(R_1 + R_2) / (1 + j\omega (R_1 + R_2) C_L)] / [\text{same}] + r_o$$

$$V_x = V_T (R_1 + R_2) / [R_1 + R_2 + r_o + j\omega r_o (R_1 + R_2) C_L]$$

$$V_x = V_T \frac{1}{1 + \frac{r_o}{R_1 + R_2} + j\omega r_o C_L}$$

simplify: $r_o \ll (R_1 + R_2) \therefore \frac{r_o}{R_1 + R_2} \approx 0$

$$V_x \approx V_T \frac{1}{1 + j\omega r_o C_L}$$

$$V_n \approx V_T \left(\frac{R_1}{R_1 + R_2} \right) \left(\frac{1}{1 + j\omega r_o C_L} \right)$$

$$\beta = \frac{-V_o}{V_T} = \frac{V_n}{V_T} = \left(\frac{R_1}{R_1 + R_2} \right) \left(\frac{1}{1 + j\omega r_o C_L} \right)$$

let $f_0 = 1/2\pi r_o C_L$

$$\beta = \frac{R_1}{R_1 + R_2} \frac{1}{1 + j f/f_0}$$

Cont $\rightarrow$

① Cont.

assemble $a(jf)$. No f_b given! Two options:

a) assume a low f_b . for example, $f_b = 10\text{Hz}$
(selected 10Hz since $f_t = 70\text{MHz}$, clean division)

$$a_0 f_b = f_t \rightarrow a_0 = 1 \times 10^6$$

$$a(jf) = 1 \times 10^6 / (1 + jf/10\text{Hz}) \text{ V/V}$$

b) assume f_b exists at 0, which causes $a(jf)$ to be of the form:

$$a(jf) = a_0 / jf, \text{ then } |a(jf_t)| = 1 = a_0 / 10\text{MHz}$$
$$a_0 = 10 \times 10^6 \text{ V/Hz/V}$$

$$a(jf) = 10 \times 10^6 / jf \text{ V/V}$$

let's use (a)

$$T(jf) = \left[\frac{10^6}{1 + jf/10} \right] \left[\frac{R_1}{R_1 + R_2} \right] \left[\frac{1}{1 + jf/R_b} \right]$$

Reminder

$$f_b = 1 / (2\pi R_b C_L)$$
$$= 1 / (471 C_L)$$

Need $T(jf)$ when angle is $45^\circ - 180^\circ = -135^\circ$

$$\angle T(jf_x) = -135^\circ$$

$$\text{and need } |T(jf_x)| = 1$$

$\rightarrow$ 2 equations, 2 unknowns (f, C_L)

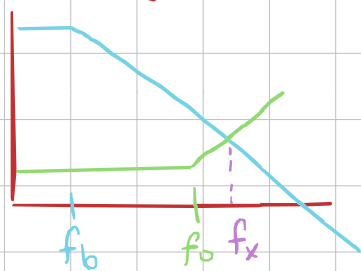
$$\angle T(jf_x) = -\arctan(f/10) - \arctan(f/R_b) = -135^\circ$$

Since we are looking for $|T(jf_x)| = 1$ & a maximum C_L before instability, then we know $f_x \gg 10\text{Hz}$

$$-135 = -90 - \arctan(f/R_b)$$

$$\arctan(f/R_b) = 45^\circ$$

$$f/R_b = 1 \rightarrow f_x = f_b!$$



$$|T(jf_x)| = 1 = 10^6 \left(\frac{R_1}{R_1 + R_2} \right) / \sqrt{1 + f_x^2/10^2} \sqrt{1 + f_x^2/R_b^2}$$

since $f_x \gg 10$, then $1 + f_x^2/10^2 \approx f_x^2/10^2$

also, since $f_x = f_b$, then $1 + f_x^2/R_b^2 = 2$

$$1 = 10^6 \left(\frac{R_1}{R_1 + R_2} \right) / \sqrt{2} \sqrt{f_x^2/10^2} = \frac{10^7}{\sqrt{2}} \left(\frac{R_1}{R_1 + R_2} \right) / f_x$$

$$f_x = 1 / (471 C_L) = \frac{10^7}{\sqrt{2}} \left(\frac{R_1}{R_1 + R_2} \right)$$

$$C_L = 3 \times 10^{-10} (1 + R_2/R_1)$$

$$a) R_1 = R_2 = 20\text{k}\Omega \rightarrow C_L = 600\text{pF}$$

$$b) R_1 = R_2 = 200\text{k}\Omega \rightarrow C_L = 600\text{pF}$$

$$c) R_1 = 2\text{k}\Omega, R_2 = 18\text{k}\Omega \rightarrow C_L = 3\text{nF}$$

$$d) R_1 = \infty, R_2 = 0\Omega \rightarrow C_L = 300\text{pF}$$

$$f_x = 3.5\text{MHz}$$

$$f_x = 3.5\text{MHz}$$

$$f_x = 708\text{kHz}$$

$$f_x = 7.08\text{MHz}$$

② See plots on following pages.

Note that both at each cursor point:

$$|T| \approx 1 \text{ (0dB)}$$

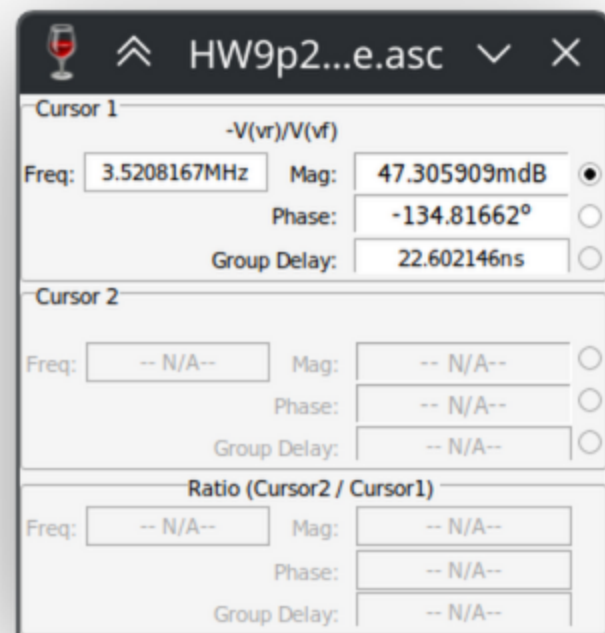
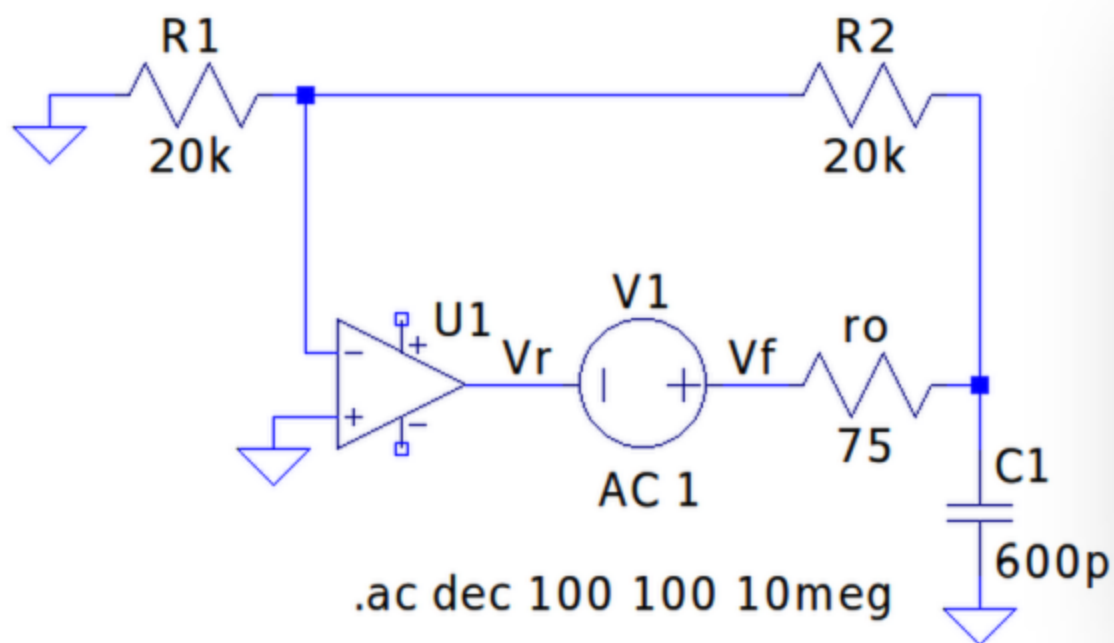
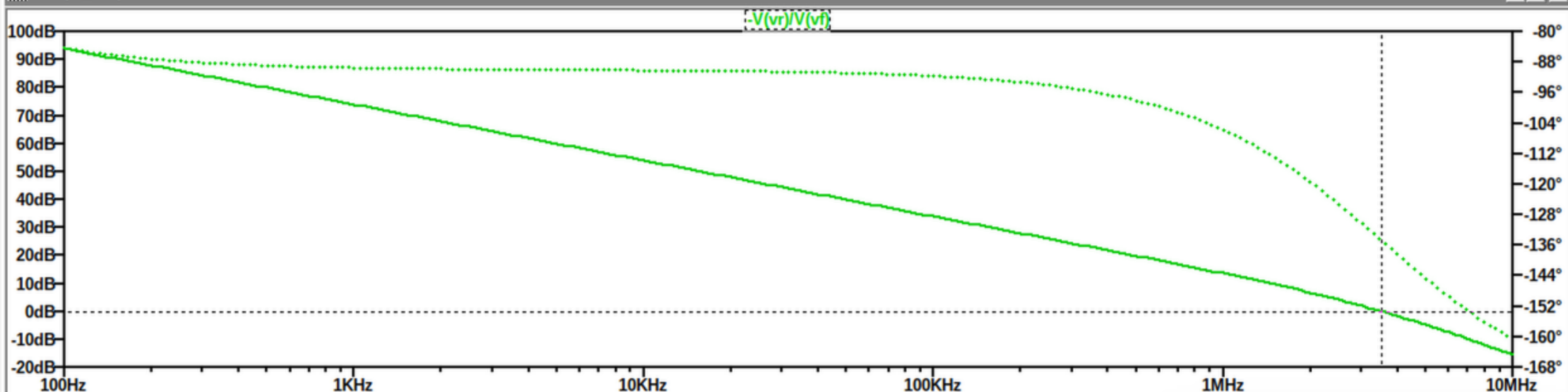
$$\angle T \approx -135^\circ (\phi_m \approx 45^\circ)$$

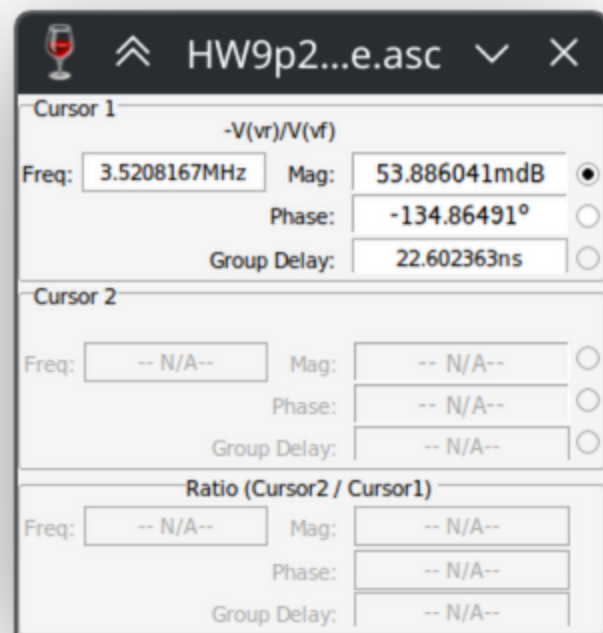
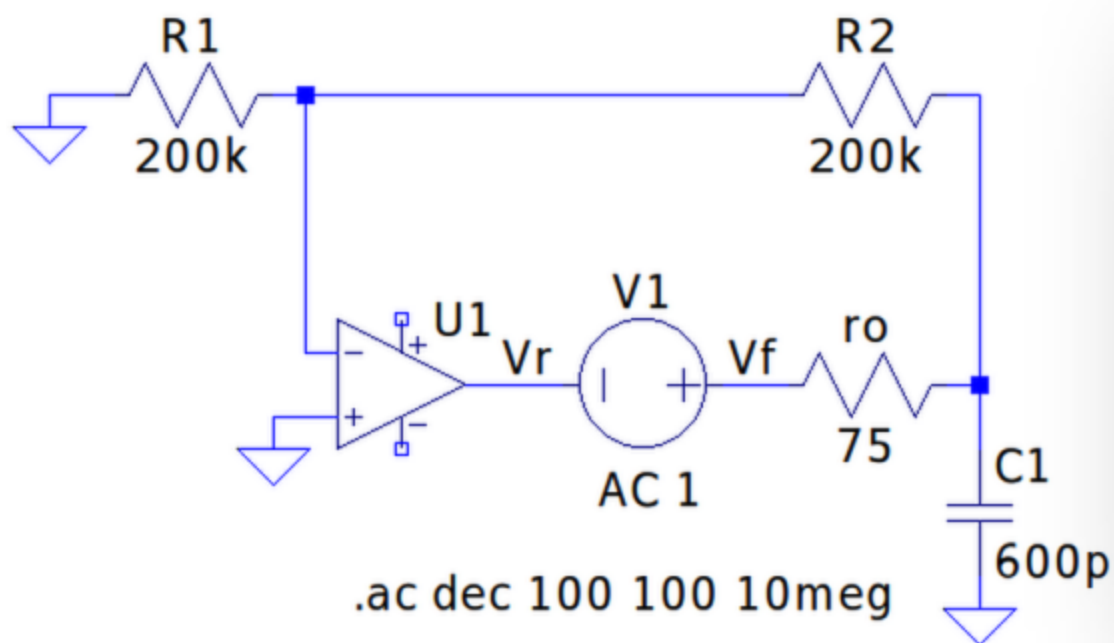
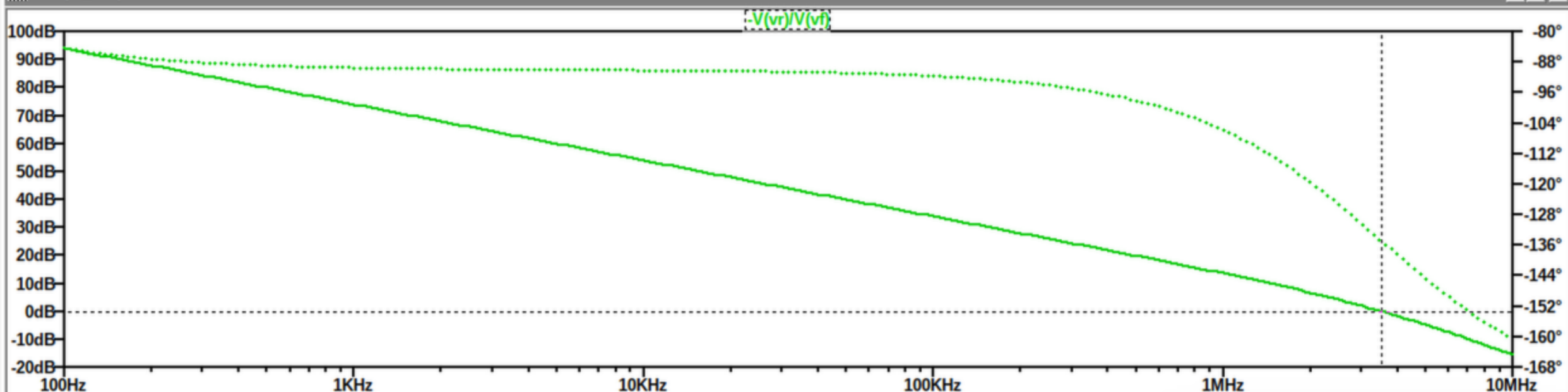
$f_x, C_L, R_1 \approx R_2$ are matched to solutions
on previous page.

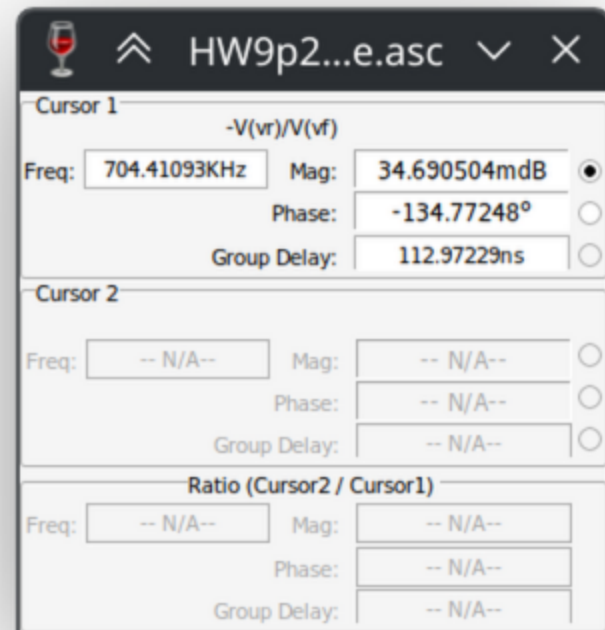
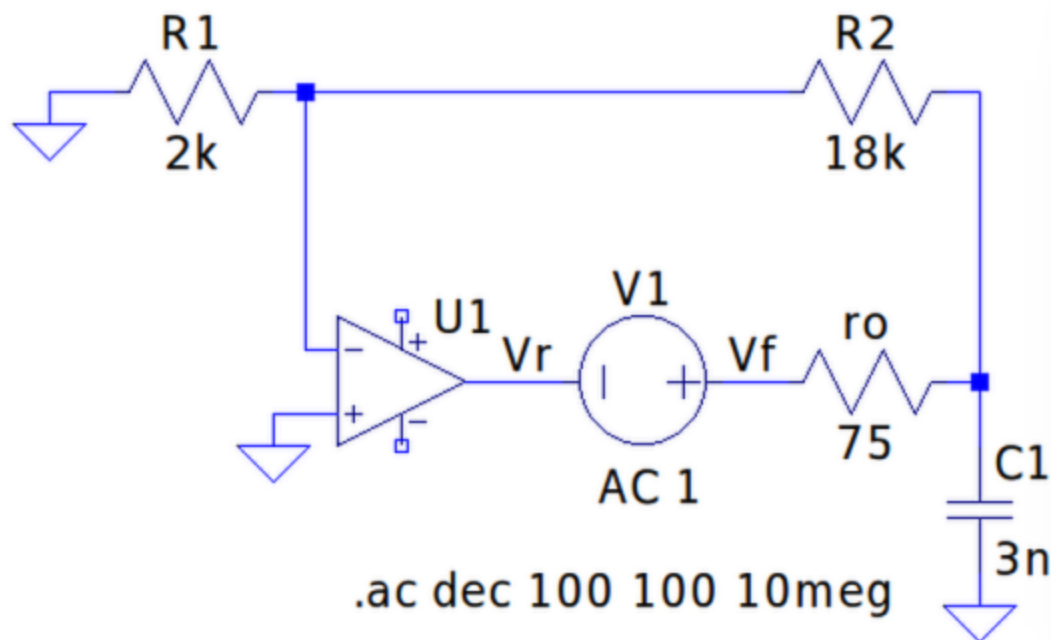
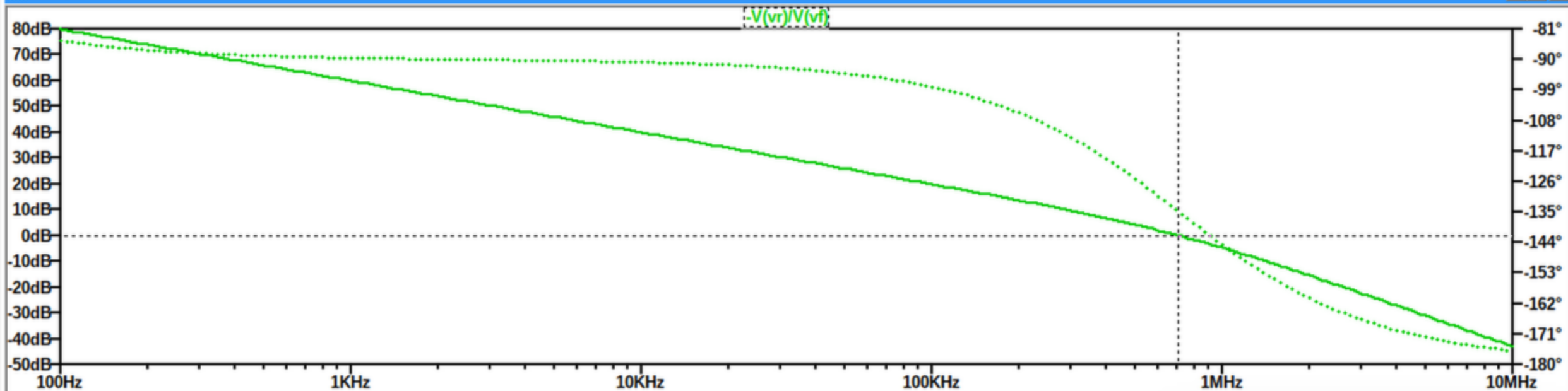
T measured using Middlebrook Voltage Injection Only

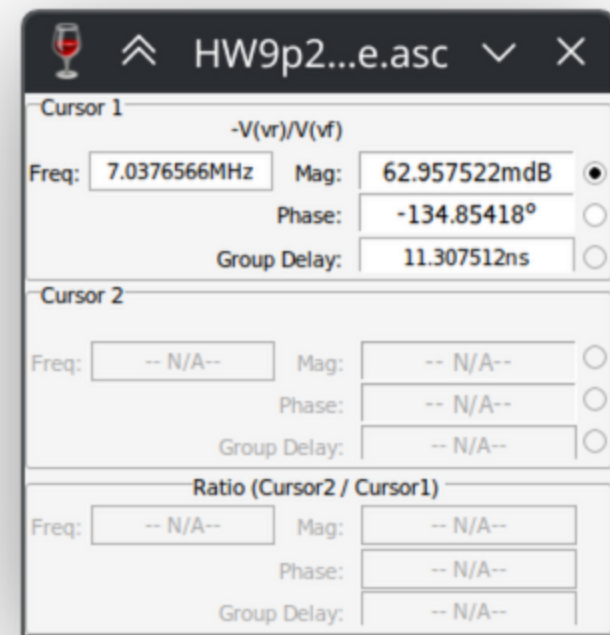
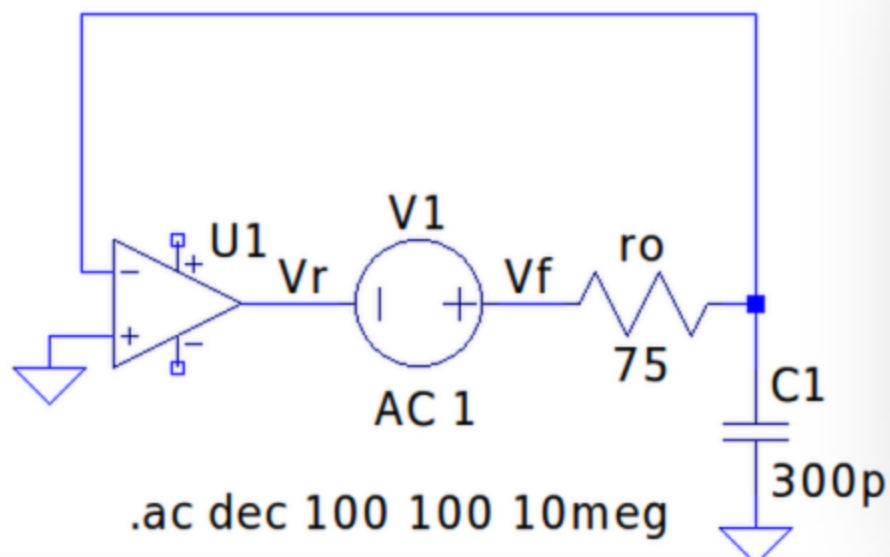
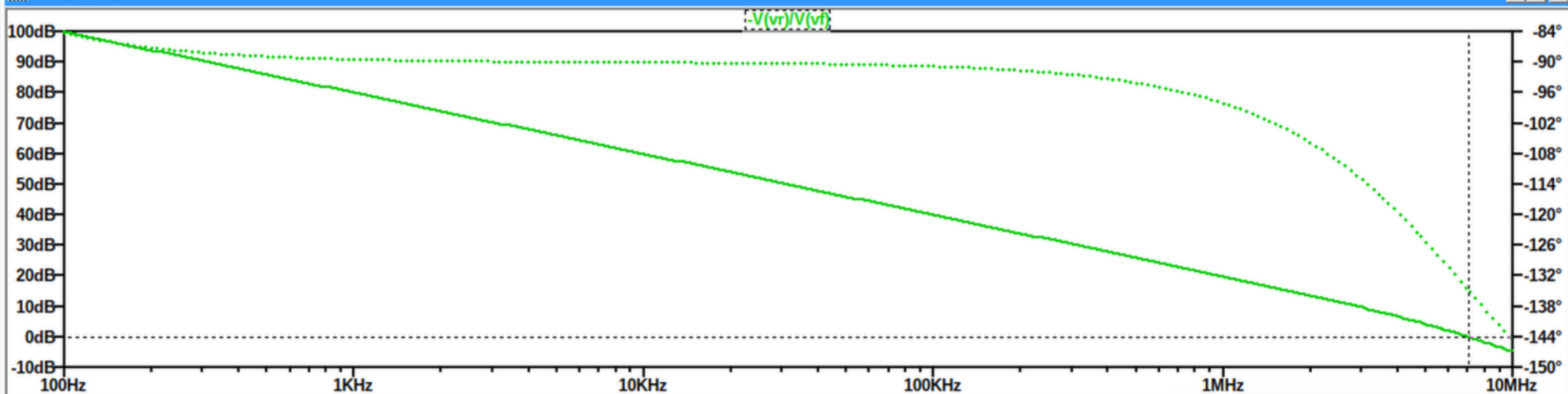
since $z_r \ll z_f$

↑ op amp output is ideal, $z_r = 0$









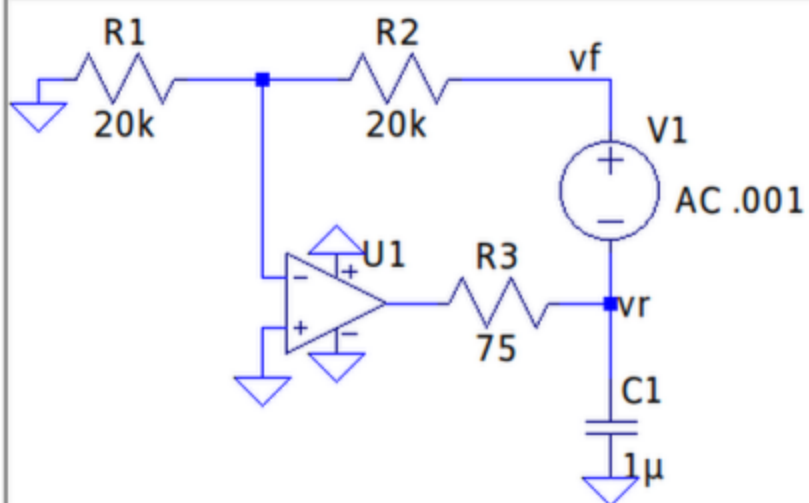
③ LTspice solutions follow on the next pages.

for each: original $T(-v_r/v_f)$ and compensated $T(-v_{r2}/v_{f2})$ are both shown.

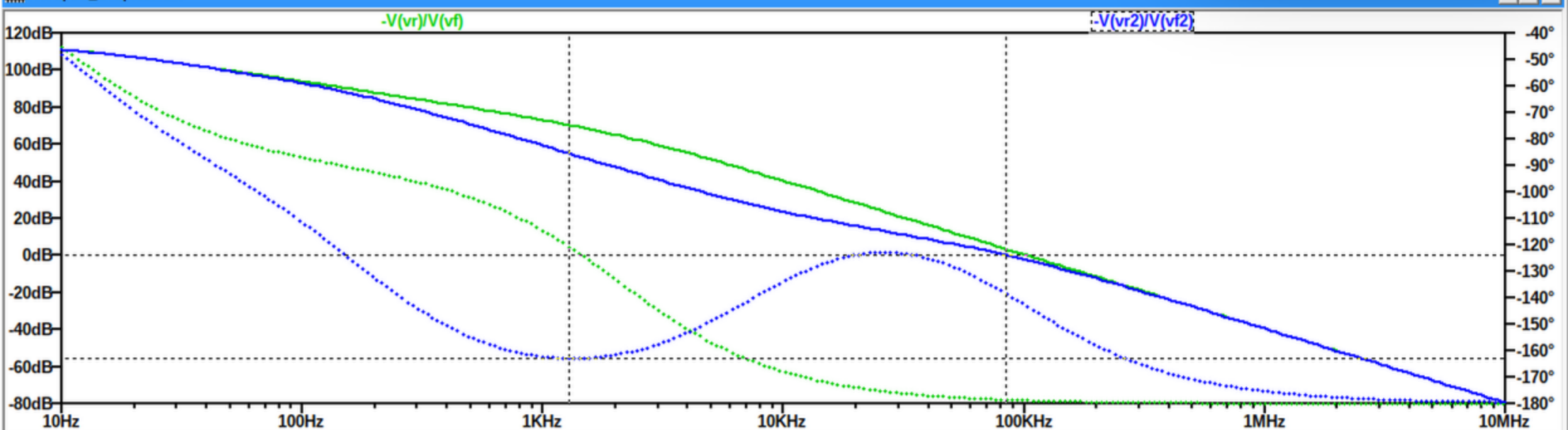
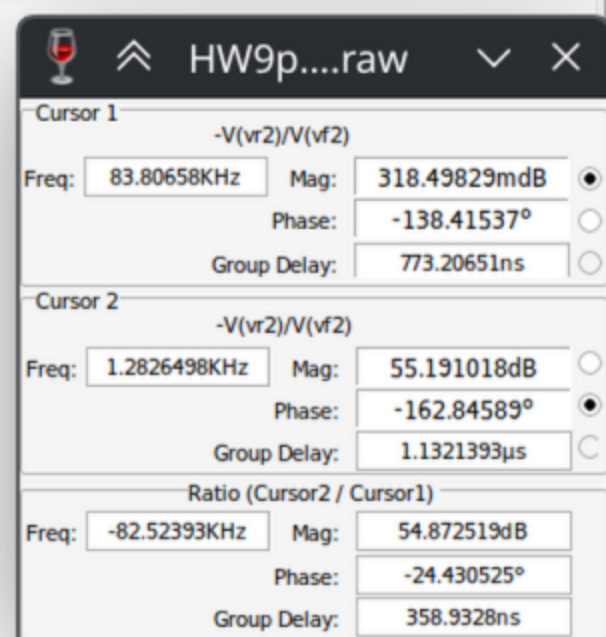
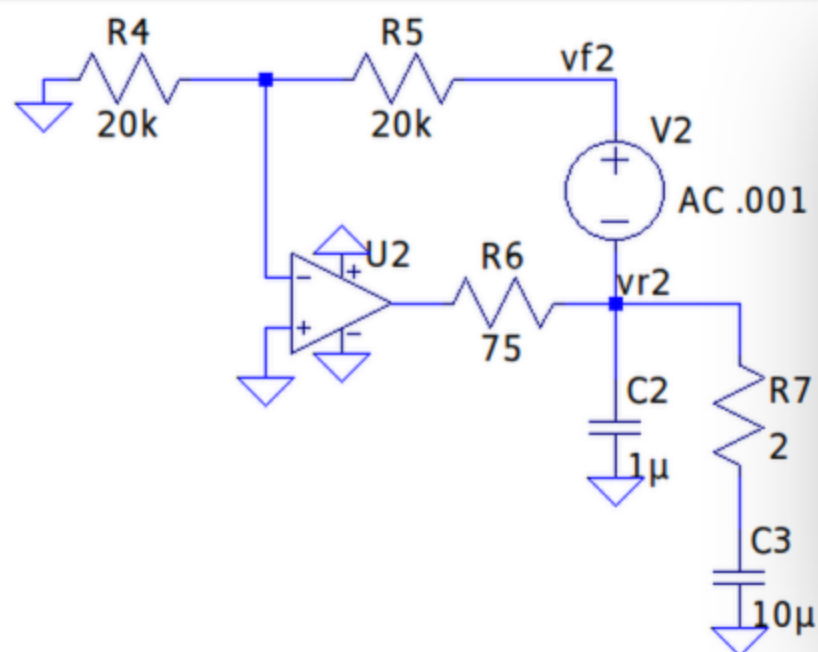
a) snubber configured with $R=2\Omega$ & $C=10\text{ nF}$
Cursor 1 marks the stability criterion, $\phi_m \approx 45^\circ$
where $f_x \approx 84\text{ kHz}$
but! note that the phase is lower when $|T| > 1$, marked by cursor 2, the minimum phase is $\approx -163^\circ$ at 1.3 kHz . This will cause significant gain peaking and ringing even though $\phi_m \approx 45^\circ$!

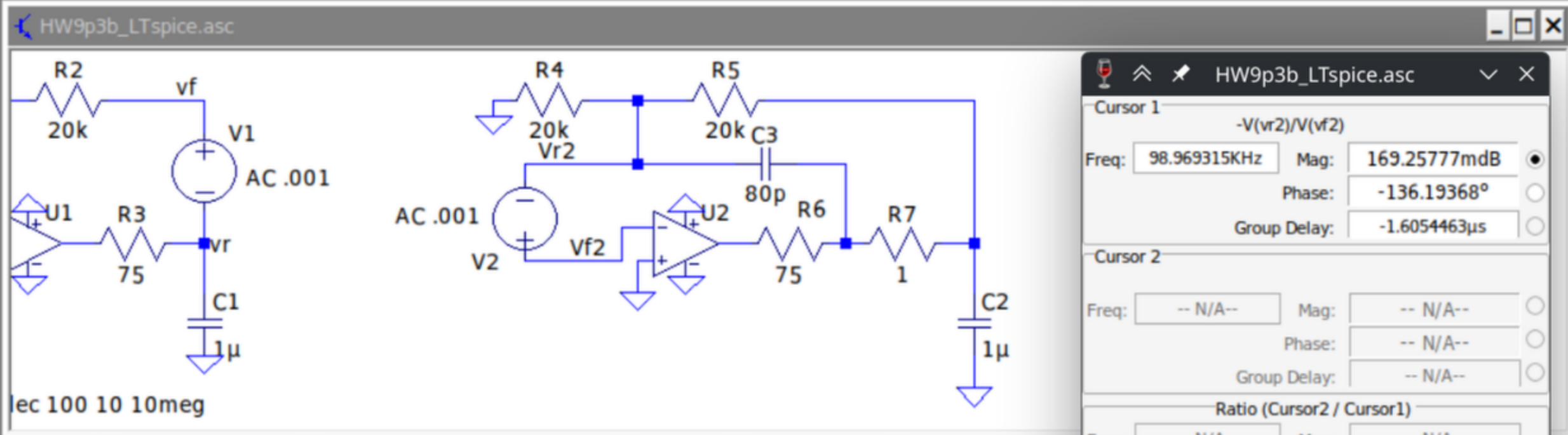
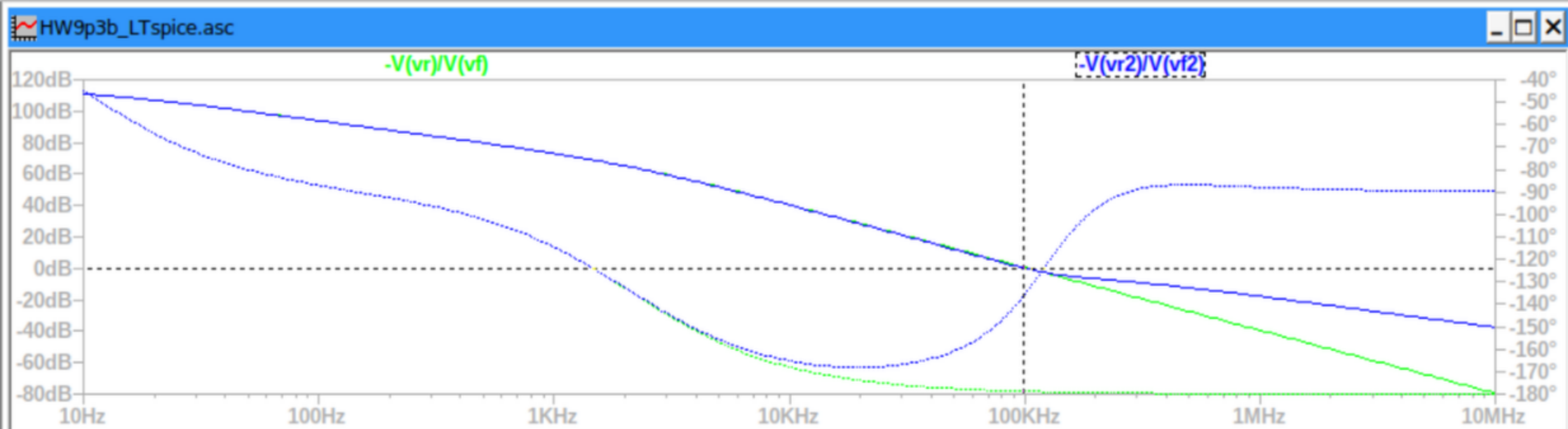
b) feedback capacitor configured with $C_f = 80\text{ pF}$ & $R_s = 1\Omega$
with $f_x \approx 99\text{ kHz}$ & $\phi_m \approx 45^\circ$
In this case, we have a higher f_x but phase is slightly worse, with a minimum of $\approx -168^\circ$ at 19.7 kHz .

Note: Voltage injection location was changed such that the feedback was captured properly (and not split by new capacitor).



.ac dec 100 10 10meg

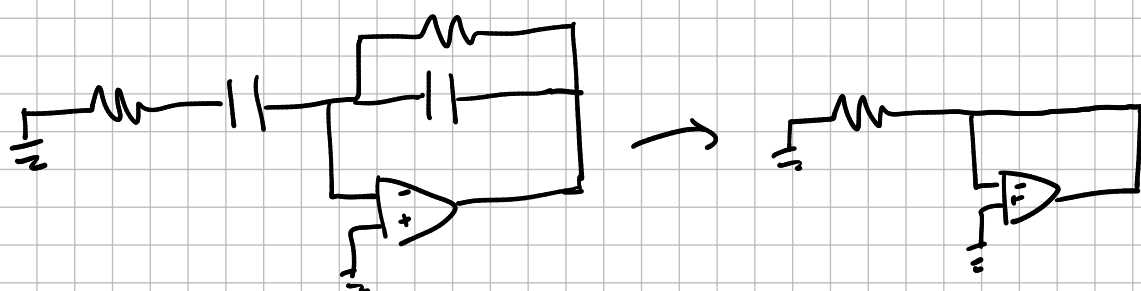




8.50

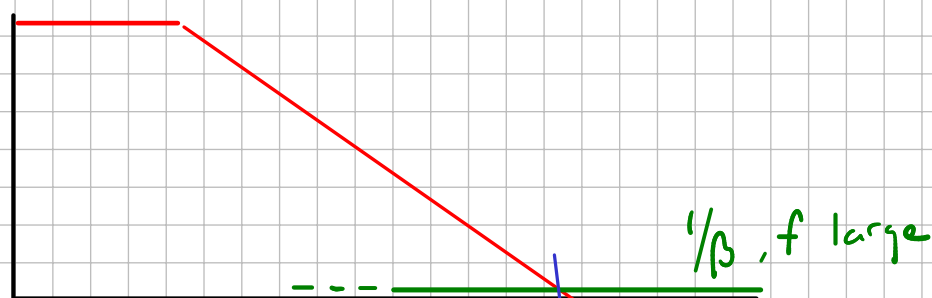
a)

Simple way: using assumption that $f_p, f_z \ll f_x$, then frequency is high enough to consider capacitors as shorts.



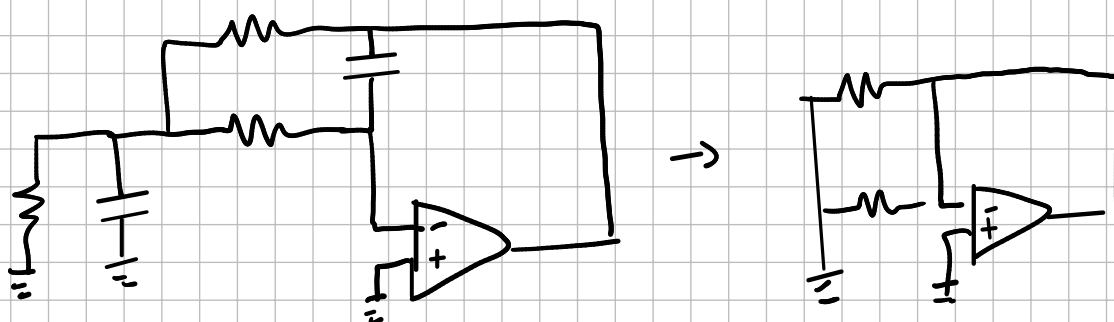
$$\beta = -\frac{V_o}{V_r}, \quad V_o = -V_n, \quad V_n = V_r \therefore \beta = -\frac{V_r}{V_r} = 1$$

β at high frequency is a constant... and 1!



f_x
 $ROC: 0 - 20 = 20$
 $\phi_m \approx 90^\circ$
Stable

b)

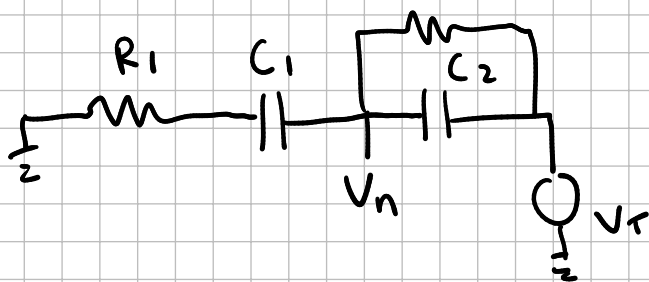


same as above!

more complicated $\rightarrow$

8.50 cont.

a) β :



$$V_D = -V_n$$

$$\beta = \frac{V_n}{V_r}$$

$$\beta = \frac{R_1 + 1/sC_1}{R_1 + 1/sC_1 + R_2} \cdot \frac{1 + sR_2C_2}{1 + sR_1C_1 + sR_2C_2}$$

$$= \frac{(1 + sR_1C_1)(1 + sR_2C_2)}{(1 + sR_1C_1)(1 + sR_2C_2) + sR_2C_1} \quad \leftarrow 2 \text{ zeros}$$

$$\quad \leftarrow 2 \text{ poles}$$

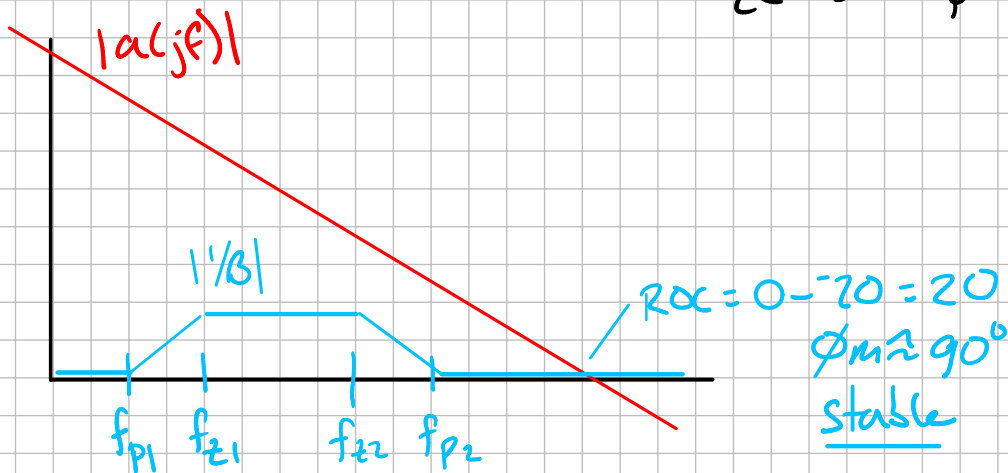
putting in #s and simplifying:

$$\beta = \frac{(1 + jf/f_{z1})(1 + jf/f_{z2})}{(1 + jf/f_{p1})(1 + jf/f_{p2})}$$

$$f_{z1} = 16 \text{ Hz} \quad f_{p1} = 7.9 \text{ Hz}$$

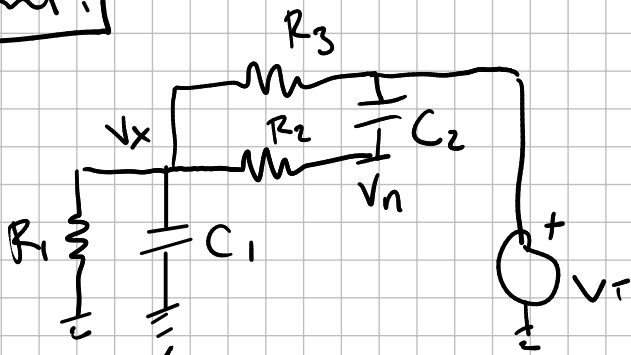
$$f_{z2} = 1.6 \text{ kHz} \quad f_{p2} = 3.2 \text{ kHz}$$

↑ these are for β ,
zeros $\leftrightarrow$ poles for $1/\beta$



8.50 cont.

b) β :



$$\begin{aligned} R_1 &= 1k & C_1 &= 100n \\ R_2 &= 10k & C_2 &= 10n \\ R_3 &= 100k \end{aligned}$$

$$V_X = V_T \frac{R_1 / (1 + s R_1 C_1)}{R_1 / (1 + s R_1 C_1) + \frac{R_3 (R_2 + 1/s C_2)}{R_3 + R_2 + 1/s C_2}}$$

(calculator)

$$V_X = \frac{1100 (s + 909)}{(s + 7340)(s + 13760)} V_T$$

$$V_n = (V_T - V_X) \frac{R_2}{R_2 + 1/s C_2} + V_X$$

(calculator)

$$\beta = \frac{(s + 90.8)(s + 11009)}{(s + 7340)(s + 13760)} = \frac{1}{100} \frac{(1 + jf/14.5)(1 + jf/1753)}{(1 + jf/1169)(1 + jf/2191)}$$

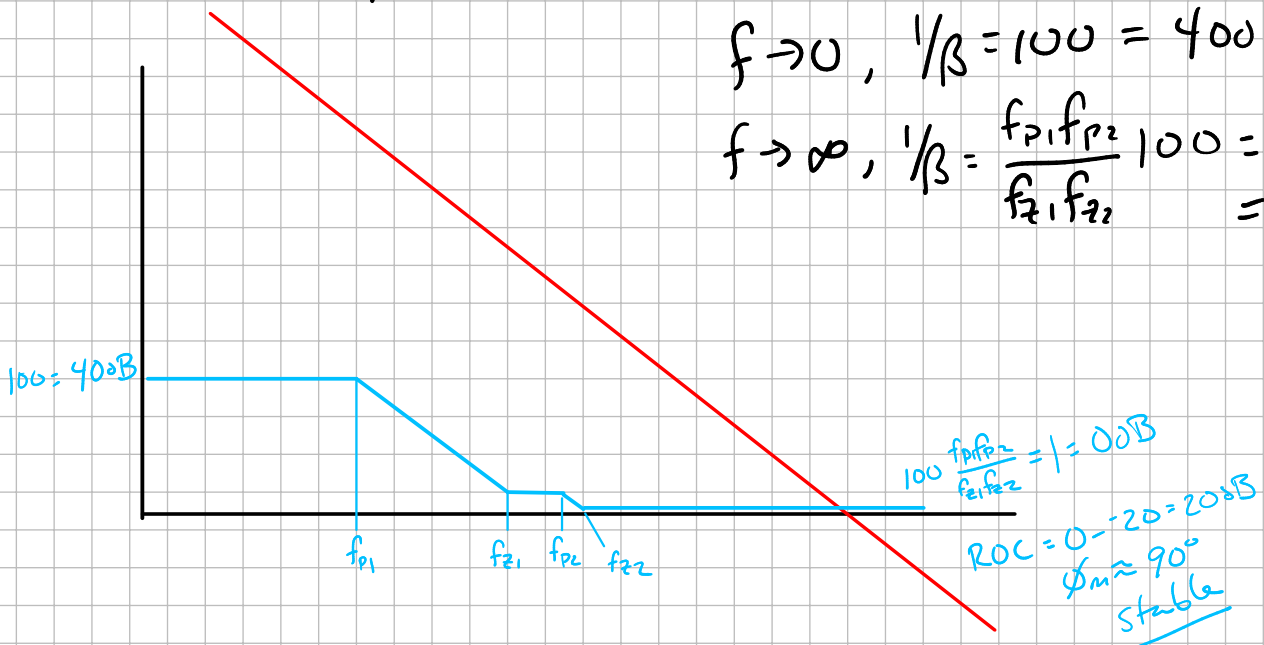
$$1/\beta = 100 \frac{(1 + jf/f_{z1})(1 + jf/f_{z2})}{(1 + jf/f_{p1})(1 + jf/f_{p2})}$$

$$f_{p1} = 14.5 \text{ Hz} \quad f_{z1} = 1169 \text{ Hz}$$

$$f_{p2} = 1753 \text{ Hz} \quad f_{z2} = 2191 \text{ Hz}$$

$$f \rightarrow 0, 1/\beta = 100 = 40 \text{ dB}$$

$$f \rightarrow \infty, 1/\beta = \frac{f_{p1} f_{p2}}{f_{z1} f_{z2}} 100 = 1457 = 63.3 \text{ dB}$$



a)

$$R_3 = 10k$$

$$1/\beta = 23 \frac{(R_c + 5)(R_c + 52.2)}{R_c(R_c + 50)} = |a(jf_2)|$$

$$|a(jf_2)| = \left| 10^5 / \left(1 + j \frac{200k}{10k} \right) \left(1 + j \frac{200k}{200k} \right) \left(1 + j \frac{200k}{2M} \right) \right| = 3514$$

solving gives: $R_C = 344 \Omega$

$$f_D = \frac{1}{2\pi R_C C_C}, \quad f_p = 0.1 f_2 = 20k$$

$$C_C = 23 nF$$

c) book says $f_B \approx f_2$.

Simulating in LTspice gives:

$$A_0 = 27.2 dB, \quad A_{-3dB} = 24.2 dB, \quad f_{-3dB} \approx 332 kHz$$